

Transmission Media

- A communication link is established by using some kind of transmission medium. In the most cases it is a pair of electric conductors.
- Communication links are also established using the media such as fiber optic cables & using electromagnetic waves on the free space.



Communication or Networking Media

1. Copper Media
2. Optical Media
3. Wireless Media

Copper Media

- Copper cable is used in almost every LAN
- Many different types of copper cable are available
 1. Twisted pair cables
 2. Coaxial Cables
- Each type has advantages and disadvantages
- Proper selection of cabling is key to efficient network operation

Twisted Pair Cables

- Oldest & still most common transmission Media.
- Consist of two insulated copper wires twisted together.
- Can be used for either analog or digital transmission.
- Twisting reduces the effects of the external interferences.
- To avoid magnetic field effect the wires are twisted.
- Common applications.
 - Telephone System ,LAN

Twisted Pair Cables

- Two types.
 - Shielded Twisted Pair Cables (STP)
 - Unshielded Twisted Pair Cables (UTP)

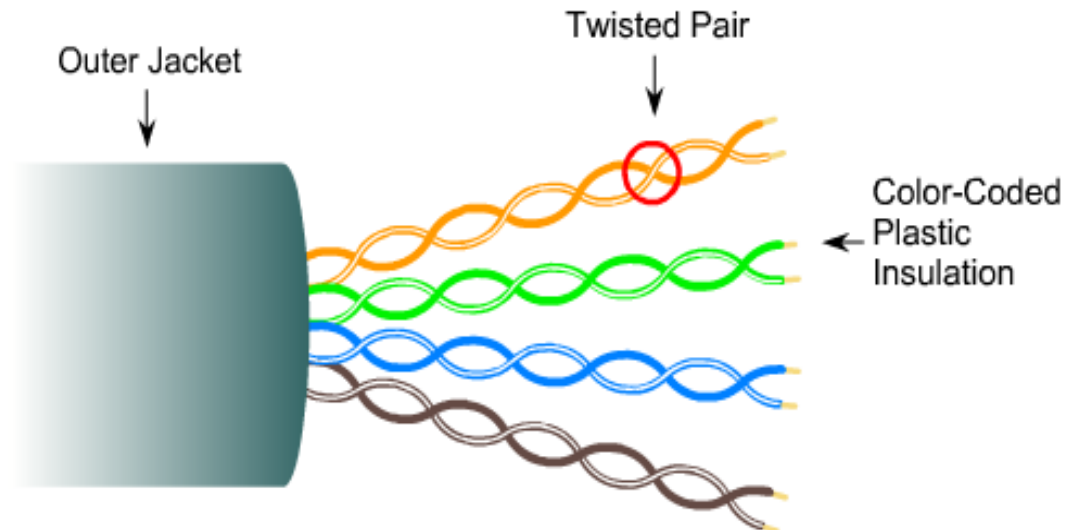
Twisted Pair Cables

- **Unshielded Twisted pair cables (UTP)**
 - Commonly used UTP are categorized as CAT3, CAT5 & CAT5e
 - Two types
 - Straight through
 - Crossover
 - Use RJ45 connector for UTP cables

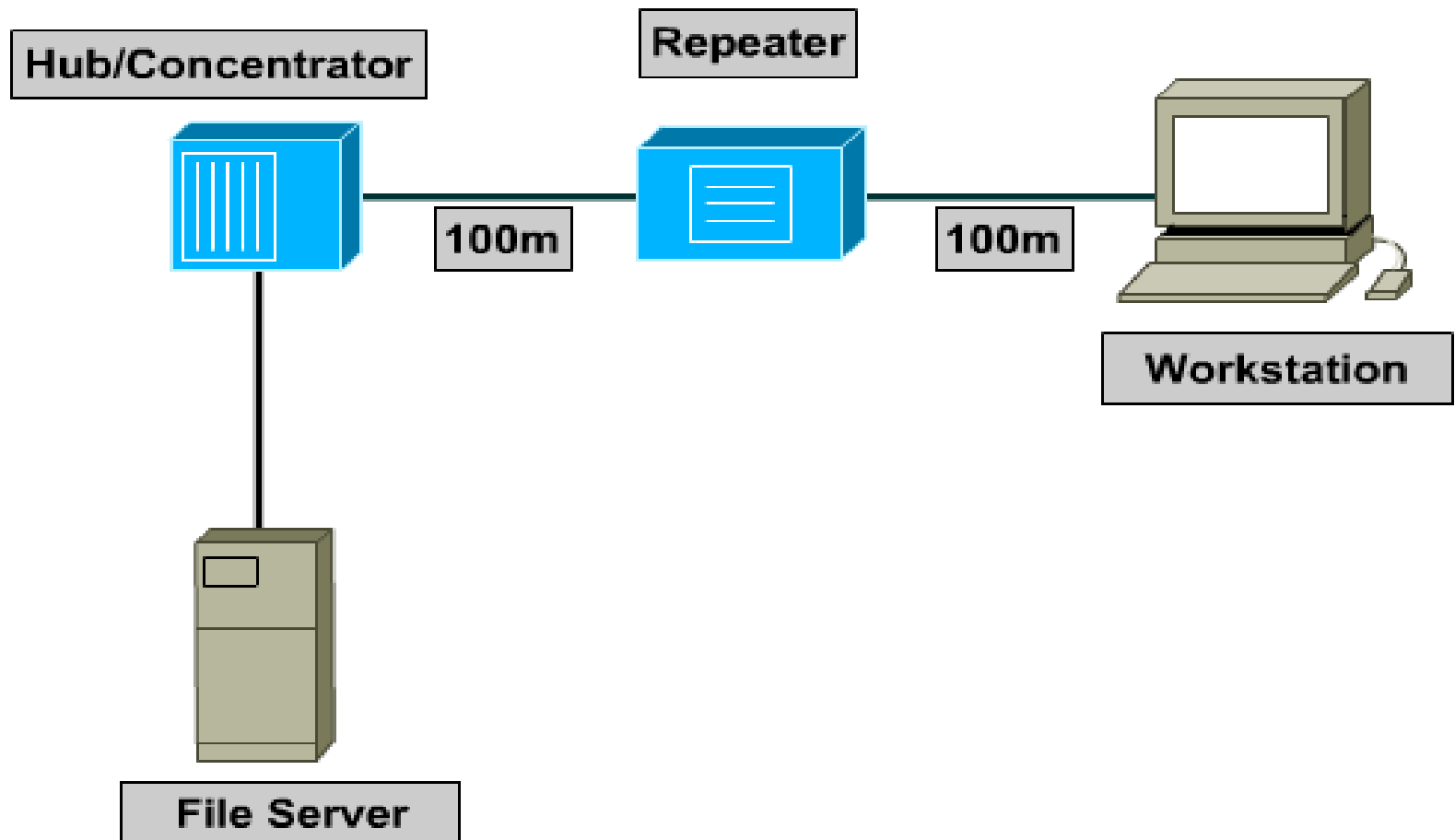
Twisted Pair Cables

- **Unshielded Twisted pair cables (UTP)**

There are 8 color wires. 4 wires for carrying voltage called Tip, 4 wires for grounded called ring



UTP Cabling



Unshielded Twisted pair cables (UTP)

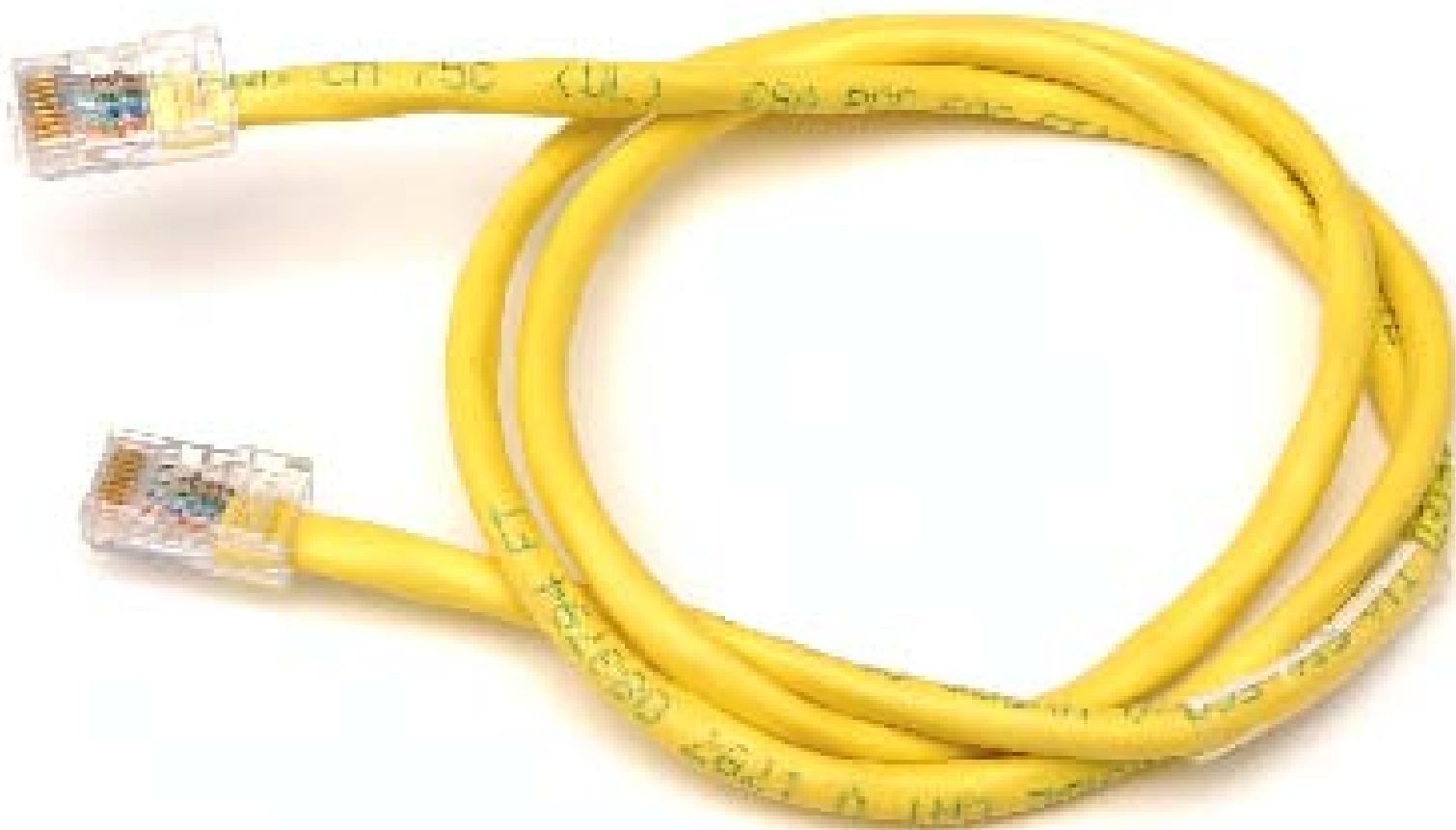
- **Connecting Different Devices – straight through cable**
 - Switch and a PC



Straight-through Cable

Pin 1	— — — — —	Pin 1
Pin 2	— — — — —	Pin 2
Pin 3	— — — — —	Pin 3
Pin 4	— — — — —	Pin 4
Pin 5	— — — — —	Pin 5
Pin 6	— — — — —	Pin 6
Pin 7	— — — — —	Pin 7
Pin 8	— — — — —	Pin 8

Straight-through Cable



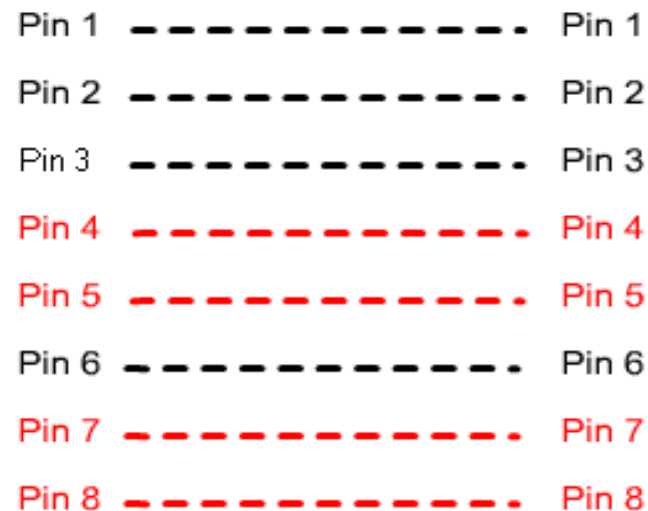
Unshielded Twisted pair cables (UTP)

- **Connecting Similar Devices – Crossover Cable**
 - Switch to switch



Crossover Cable

NIC transmits
signals on pins
1 & 2, it receives
signals on pin 3
& 6



An Ethernet (10BASE-T and 100BASE-TX) cross-connect cable has only four active wires 1, 2, 3, and 6

Crossover Cable





Use straight-through cables for the following connections:

- Switch to router
- Switch to PC or server
- Hub to PC or server

Use crossover cables for the following connections:

- Switch to switch
- Switch to hub
- Hub to hub
- Router to router
- PC to PC
- Router to PC

Characteristics of UTP

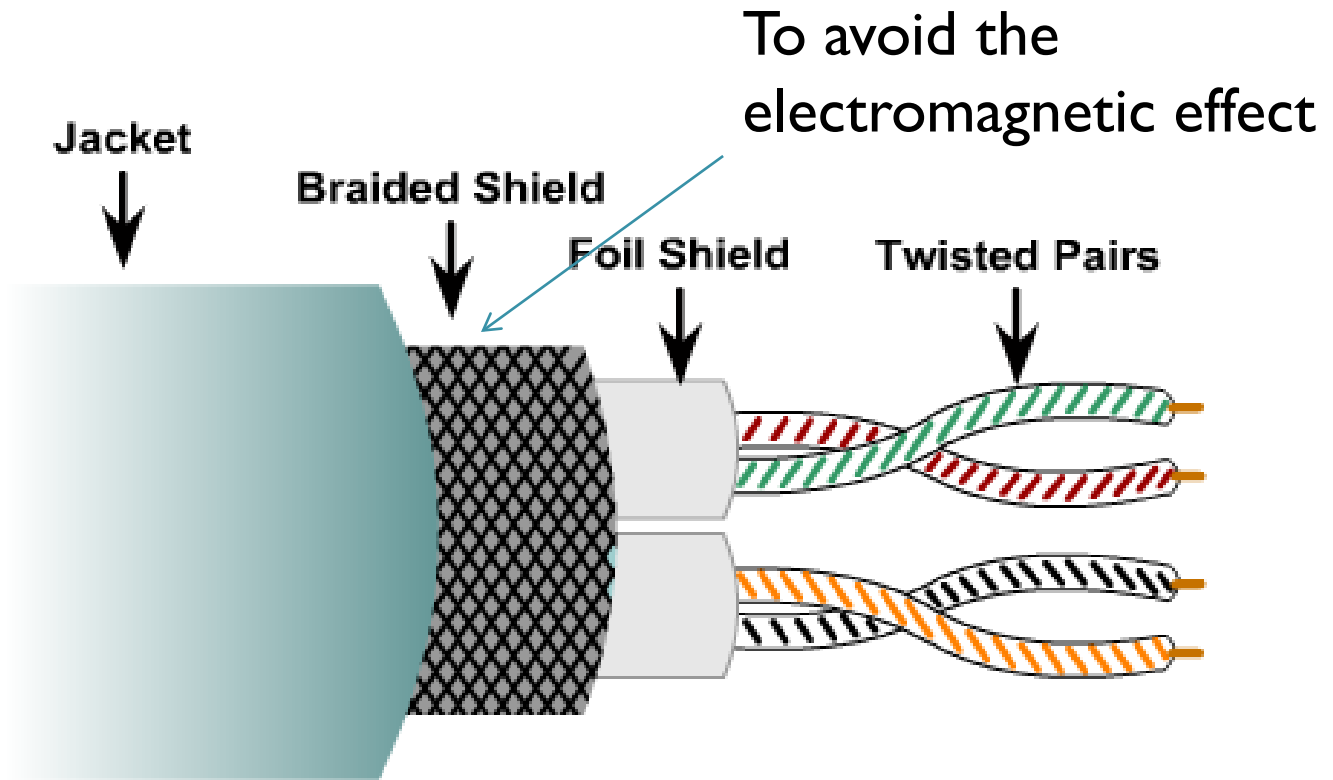
- Speed 10,100,1000 Mbps.
- Least expensive.
- Maximum cable length is 100 m.
- Media connector size is small.

Categories of Unshielded Twisted Pair

Category	Speed	Use
1	1 Mbps	Voice Only (Telephone Wire)
2	4 Mbps	LocalTalk & Telephone (Rarely used)
3	16 Mbps	10BaseT Ethernet
4	20 Mbps	Token Ring (Rarely used)
5	100 Mbps (2 pair)	100BaseT Ethernet
	1000 Mbps (4 pair)	Gigabit Ethernet
5e	1,000 Mbps	Gigabit Ethernet
6	10,000 Mbps	Gigabit Ethernet

Twisted Pair Cables

- **Shielded Twisted Pair Cables**



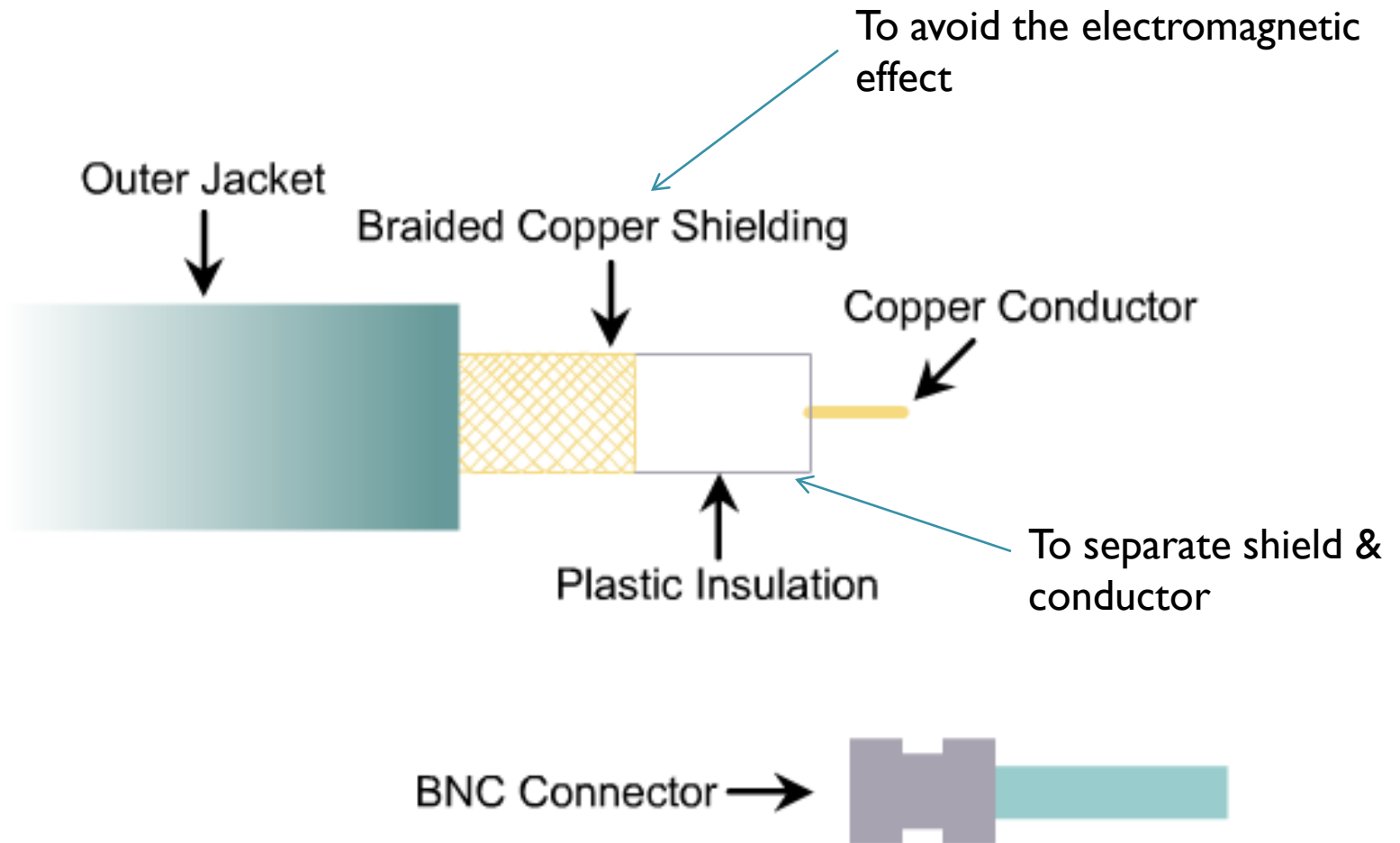
Characteristics of STP

- Speed 10-100 Mbps.
- Moderately expensive .
- Maximum cable length is 100 m.
- Similar in construction of the UTP.

More...

- Although UTP cable is the least expensive cable, it may be susceptible to radio and electrical frequency interference (it should not be too close to electric motors, fluorescent lights, etc.).
- If you must place cable in environments with lots of potential interference, or if you must place cable in extremely sensitive environments that may be susceptible to the electrical current in the UTP, shielded twisted pair may be the solution.

2. Coaxial cable



Coaxial cable

- has a single copper conductor at its center.
- A plastic layer provides insulation between the center conductor and a braided metal shield.
- The metal shield helps to block any outside interference .
- difficult to install.
- support greater cable lengths between network devices than twisted pair cable.

Coaxial cable

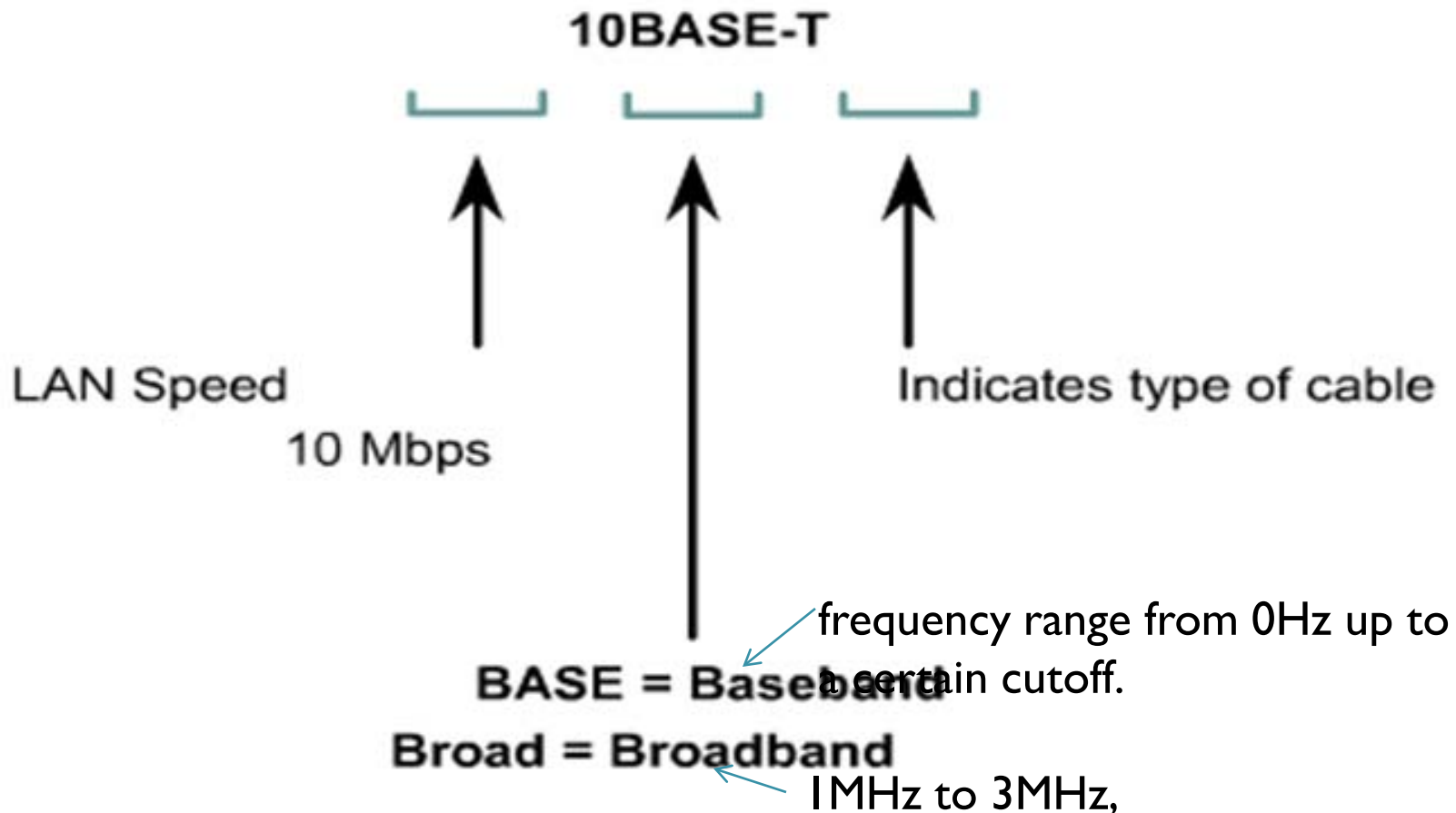
The two types of coaxial

- Thin coaxial cable is also referred to as thinnet.
 - 10Base2 refers to the specifications for thin coaxial cable carrying Ethernet signals.
- Thick coaxial cable is also referred to as thicknet.
 - 10Base5 refers to the specifications for thick coaxial cable carrying Ethernet signals.

Characteristics of Co –axial Cables

- Speed 10-100 Mbps.
- Inexpensive.
- Maximum cable length is 500 m.
- Can be run longer distances than shielded twisted pair(STP), unshielded twisted pair(UTP) without the need for repeaters.
- Media & connector size is Medium

Cable Specifications



Ethernet Cable Summery

Specification	Cable Type
10BaseT	Unshielded Twisted Pair
10Base2	Thin Coaxial
10Base5	Thick Coaxial
100BaseT	Unshielded Twisted Pair
100BaseFX	Fiber Optic
100BaseBX	Single mode Fiber
100BaseSX	Multimode Fiber
1000BaseT	Unshielded Twisted Pair
1000BaseFX	Fiber Optic
1000BaseBX	Single mode Fiber
1000BaseSX	Multimode Fiber

Cable Specifications

- Cables have different specifications and expectations.
- Important considerations related to performance are as follows:
 - ✓ Speeds for data transmission
 - ✓ Transmission be analog or digital
 - ✓ How far signal can travel before attenuation

e.g.

- 10BASE-T
- 10BASE5
- 10BASE2
- 100BASE-T

Cable Specifications (contd.)

Typical Media	Maximum Theoretical Bandwidth	Maximum Theoretical Distance
50-Ohm Coaxial Cable (10BASE2 Ethernet; Thinnet)	10 Mbps	185 m
50-Ohm Coaxial Cable (10BASE5 Ethernet; Thicknet)	10 Mbps	500 m
Category 5 Unshielded Twisted Pair (UTP) (10BASE-T Ethernet)	10 Mbps	100 m
Category 5 Unshielded Twisted Pair (UTP) (100BASE-TX Ethernet)	100 Mbps	100 m
Category 5 Unshielded Twisted Pair (UTP) (1000BASE-TX Ethernet)	1000 Mbps	100 m
Multimode Optical Fiber (62.5/125 μ m) (100BASE-FX Ethernet)	100 Mbps	220 m
Multimode Optical Fiber (62.5/125 μ m) (100BASE-FX Ethernet)	1000 Mbps	220 m
Multimode Optical Fiber (50/125 μ m) (1000BASE-SX Ethernet)	1000 Mbps	550 m
Multimode Optical Fiber (50/125 μ m) (1000BASE-SX Ethernet)	1000 Mbps	550 m
Singlemode Optical Fiber (9/125 μ m) (1000BASE-LX Ethernet)	1000 Mbps	5000 m

Ethernet Media and Connector Requirement

	10BASE2	10BASE5	10BASE-T	100BASE-TX	100BASE-FX	1000BASE-CX	1000BASE-T
Media	50-ohm coaxial (Thinnet)	50-ohm coaxial (Thicknet)	EIA/TIA Category 3, 4, 5 UTP, two pair	EIA/TIA Category 5 UTP, two pair	62.5/125 multimode fiber	STP	EIA/TIA Category 5 UTP, four pair
Maximum Segment Length	185 m (608.94 feet)	500 m (1640.4 feet)	100 m (328 feet)	100 m (328 feet)	400 m (1312.3 feet)	25 m (82 feet)	100 m (328 feet)
Topology	Bus	Bus	Star	Star	Star	Star	Star
Connector	BNC	Attachment unit interface (AUI)	ISO 8877 (RJ-45)	ISO 8877 (RJ-45)		ISO 8877 (RJ-45)	ISO 8877 (RJ-45)

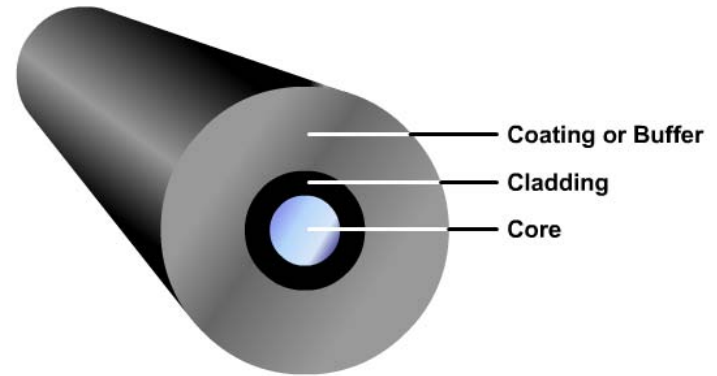
Fiber Optics

- Fiber optic cable is a completely different type of network medium than twisted-pair or coaxial cable.
- Instead of carrying signals over copper conductors in the form of electrical voltages, fiber optic cables transmit pulses of light over a glass or plastic filament.

Fiber Optics

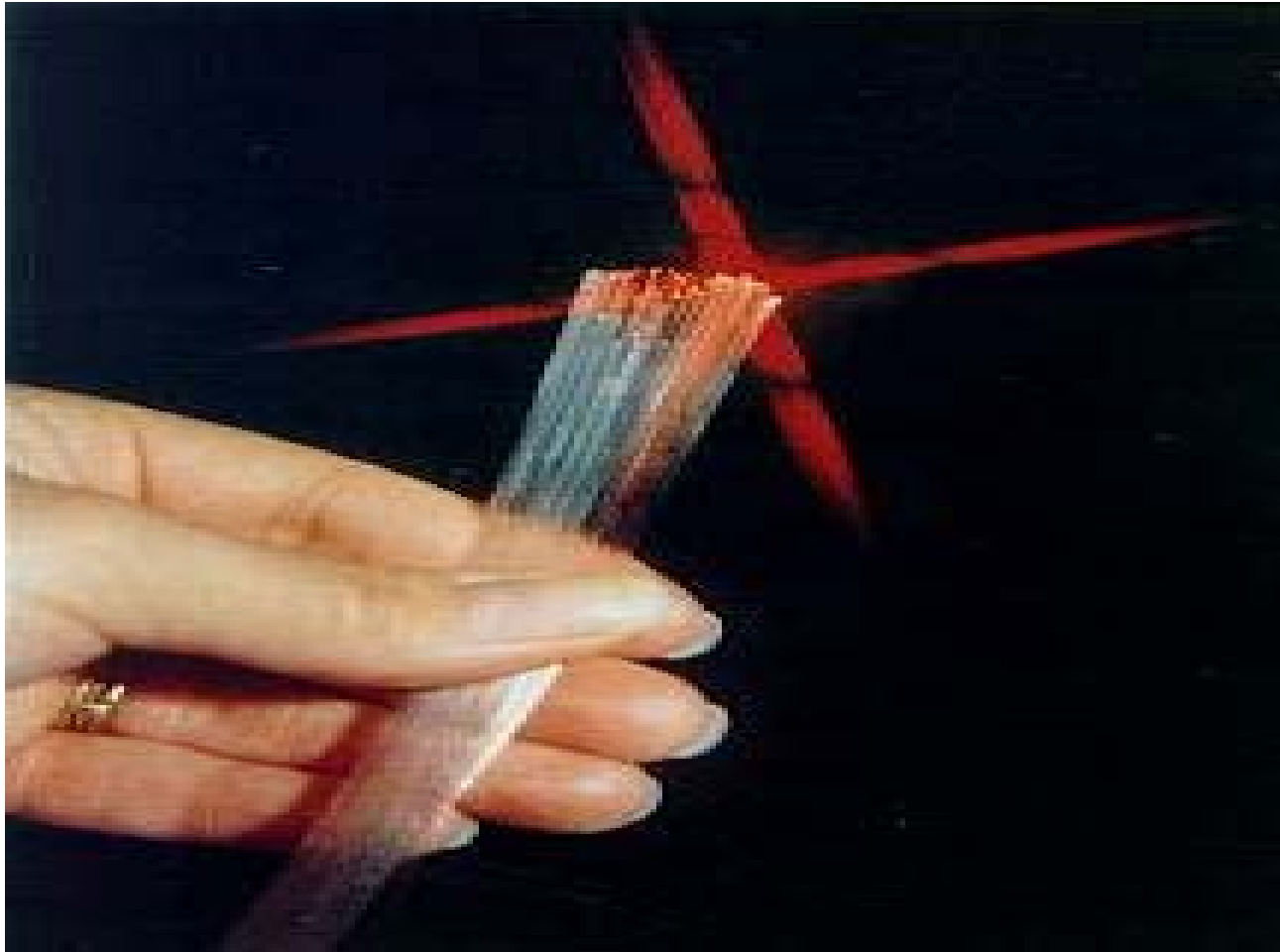
- Optical fiber is the most frequently used medium for the longer, high bandwidth, point-to-point transmissions required on LAN backbones and on WANs .
- consists of a center glass core surrounded by several layers of protective .
- has the ability to transmit signals over much longer distances (upto 2Km)than coaxial and twisted pair.

Fiber Optics



- has the capability to carry information at vastly greater speeds.(1000 Mbps)
- Consist of extremely thin cylinder of glasses called core surrounded by claddy.
- More lightly expensive than Cu cables
- Two fibers. one to transmit & one to receive.
- High secure from outside interference such as radio transmitter, arc welders, fluorescent light & other sources of electrical noises.

Fiber optic

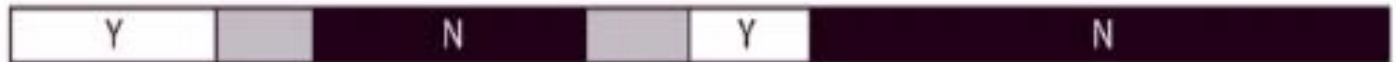


Wireless Media

- Transmission of waves take place in the electromagnetic (EM) spectrum.
- The carrier frequency of the data is expressed in cycles per second called hertz(Hz).
- Low frequency signals can travel for long distances through many obstacles but can not carry a high bandwidth of data
- high frequency signals can travel for shorter distances through few obstacles and carry a narrow bandwidth.
- Also the noise effect on the signal is inversely proportional to the power of the radio transmitter.

THE ELECTROMAGNETIC SPECTRUM

Penetrates
Earth
Atmosphere?



Wavelength
(meters)



About the size of...



Buildings



Humans



Honey Bee



Pinpoint



Protozoans



Molecules

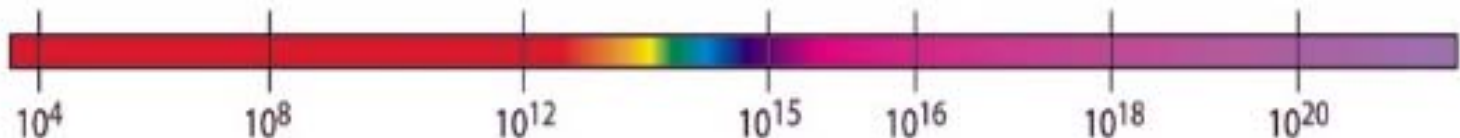


Atoms



Atomic Nuclei

Frequency
(Hz)



Wireless Media

The three broad categories of wireless media are:

- **Radio**

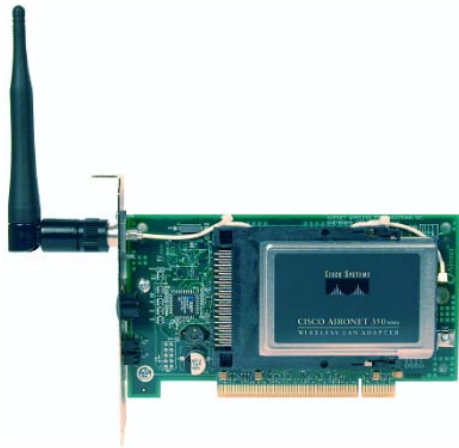
- 10 KHz to 1 GHz. It is broken into many bands including AM, FM, and VHF bands.

- **Microwave**

- **Infrared**

- The signal cannot travel through objects. Light may interfere with the signal.

Wireless Devices



internal wireless NIC for Desktop or server



PCMCIA NIC for Laptop



External USB Wireless NIC

Access Point(AP)

- ❑ To solve compatibility problems among devices, an infrastructure mode topology can be set up using an access point (AP) to act as a central hub for the WLAN.
- ❑ The AP is hard wired to the cabled LAN to provide Internet access and connectivity to the wired network.
- ❑ APs are equipped with antennae and provide wireless connectivity over a specified area

